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Precision and Efficiency in Dual-Lateral Wells: Innovative and Advanced Coiled Tubing Technology in Well Access for Acid Stimulation

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Abstract

Well intervention operations in dual-lateral wells are inherently challenging due to complex well trajectories and intricate design. This paper showcases the application of advanced technologies to simplify acid stimulation in dual-lateral open-hole wells. The focus is on tools and techniques that enable efficient lateral entry without the need for extensive mapping, driving significant operational efficiency and cost savings.

This intervention integrates real-time downhole sensors (RTDS) with the hydraulic Multilateral Entry Tool (MLET) to optimize acid stimulation workflows. The advanced sensor systems deliver continuous depth correlation, pressure monitoring, and precise tool face orientation, enabling seamless entry into the lateral windows. These technologies eliminate the time-intensive mapping process, streamlining operations and minimizing non-productive time.

The paper also includes a comparative analysis of this innovative methodology versus conventional dual-lateral acid stimulation operations, focusing on metrics such as operational time and success rates to access the wellbore laterals in acid stimulation operations.

This field application demonstrated substantial improvements in efficiency and operational outcomes. The RTDS provided actionable data, allowing for optimal tool face alignment and immediate lateral access. This capability significantly reduced operational delays and improved decision-making during the intervention. The MLET along with tool face orientation, designed for accuracy and efficiency, leverages downhole feedback to ensure precise lateral access. This approach allows for dynamic adjustments during the operation, eliminating the need for multiple iterations and enhancing overall efficiency.

The real time downhole sensors and tool face almost eliminated the time typically required for iterative mapping process with the conventional MLET; the system delivered faster results and ensured the operational scope was met without compromises. These combined technologies reduced cost, minimized operational duration and downtime, and enhanced mapping accuracy across the window.

This paper introduces a novel approach to dual-lateral acid stimulation, showcasing the synergistic potential of advanced sensor technology and the MLET with tool face orientation. The successful application offers a scalable solution for tackling the complexities of dual-lateral interventions and lateral accessibility, setting a benchmark for efficiency, safety, and comprehensive reservoir stimulation.

Introduction

The tool face orientation in well intervention operations is a critical factor that directly impacts the success of accessing lateral windows in dual-lateral wells. Accurate tool face alignment is essential for optimizing acid stimulation and ensuring efficient operations. In dual-lateral wells, where multiple lateral branches extend from a single vertical or deviated wellbore, the complexity of interventions is heightened. Operators are tasked with navigating varied geological formations and ensuring precise entry into each lateral to maximize reservoir contact and production efficiency.

Traditional methods often rely on iterative mapping processes to establish tool face orientation, which can be time-consuming and prone to errors. These delays not only increase operational costs but can also jeopardize the overall success of the intervention. Furthermore, in scenarios where multiple laterals are involved, the risk of misalignment escalates, potentially leading to non-accessible laterals or inefficient acid placement.

The tool face introduces a cutting-edge approach that leverages advanced sensor technology to enhance tool face orientation and streamline lateral access. By integrating RTDS with the hydraulic MLET, this methodology provides continuous feedback on tool position and alignment. This integration significantly reduces the need for extensive mapping, allowing operators to dynamically adjust their approach based on real-time data.

Unique scenarios in dual-lateral interventions illustrate the advantages of this technology. For instance, in cases where lateral windows are closely spaced, traditional mapping may struggle to distinguish between them, leading to costly errors. However, the precise data provided by real-time sensors allows operators to navigate these tight tolerances effectively. Additionally, when dealing with challenging formations, such as those exhibiting high levels of uncertainty or unexpected pressure variations, the ability to quickly adjust tool face orientation based on live feedback is invaluable.

The ensuing sections will explore the technological innovations implemented, including the application of the MLET and the role of real-time sensors in enhancing tool face accuracy. We will also conduct comparative analysis with conventional practices, demonstrating the resultant improvements in operational efficiency and effectiveness in dual-lateral acid stimulation. This novel approach not only sets a new benchmark for dual-lateral interventions but also underscores the transformative potential of integrating advanced technologies in well operations.

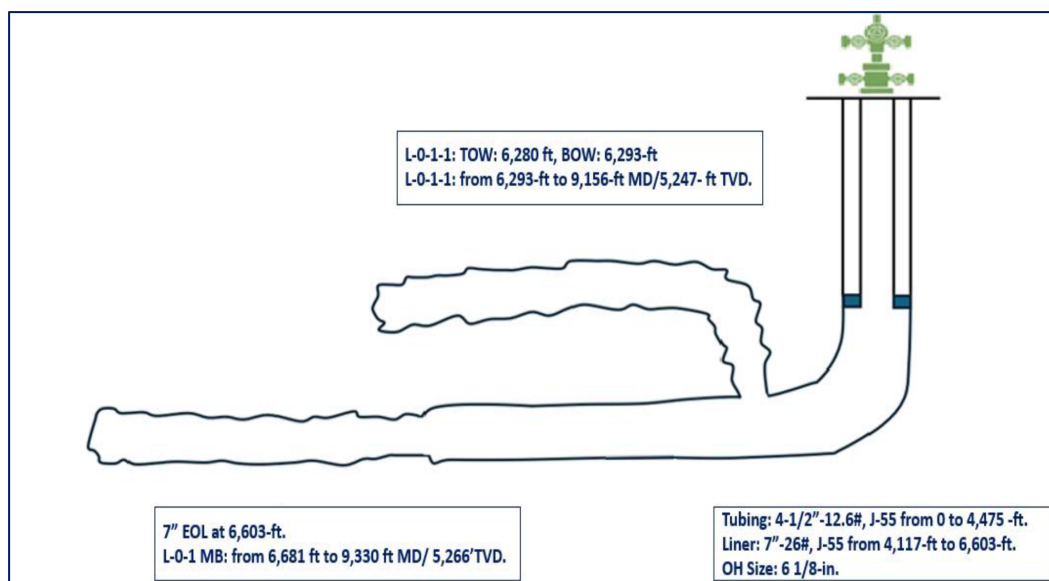


Figure 1—Wellbore Schematics

Methodology

Advanced Sensor Integration

The integration of RTDS into the MLET enables operators to receive immediate feedback on tool face orientation and position. These sensors monitor various parameters, including pressure differentials and tool alignment, allowing for dynamic adjustments throughout the intervention process.

Real-Time Monitoring. The downhole sensors continuously monitor various parameters, including pressure differentials, temperature, casing collar locator (CCL) and tension compression. This real-time data allows operators to assess the operational environment instantly and make informed decisions during the intervention.

Tool Face Orientation. The sensors specifically designed for orientation provide high-accuracy readings of the tool's angle relative to the wellbore. This precision is critical for ensuring that the MLET is properly aligned for effective lateral access. Data from these sensors enable operators to adjust the tool face in real-time, ensuring optimal entry angles for acid stimulation.

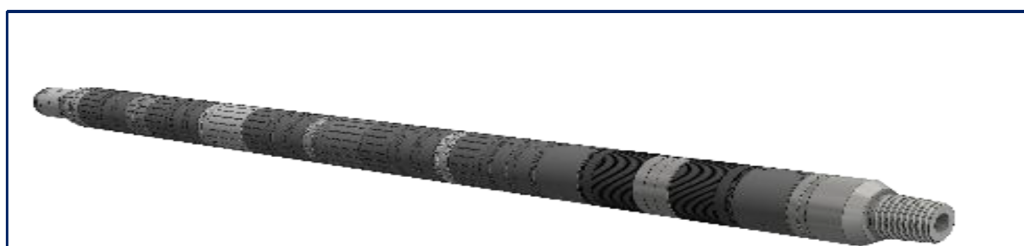


Figure 2—RTDS with Tool Face Bottom Hole Assemble (BHA)

Multilateral Entry Tool (MLET)

The MLET is designed to facilitate efficient lateral access in dual-lateral wells. It incorporates advanced features such as a bending sub that can adjust its angle based on real-time data from downhole sensors. This adaptability is crucial for navigating complex windows.

Degree of Rotation. The MLET is engineered to accommodate a rotation of up to 360° degrees from its default position by going through 12 pumping cycles (Rotating 30° degrees for each cycle). This degree of flexibility allows for precise alignment with the required entry points, ensuring optimal angles that are crucial for effective acid stimulation.

Bending Sub Mechanism. The MLET features a bending sub that enables the tool to adjust its angle dynamically based on real-time data received from the integrated sensors. This adaptability is essential for navigating various wellbore configurations and responding to unexpected geological changes during the intervention process.

Guidance System. The MLET is equipped with an advanced guidance system that utilizes data from real-time sensors to determine the exact orientation of the tool face. By continuously monitoring the tool's position, the system can make necessary adjustments to maintain alignment, enhancing the likelihood of successful access.

Operational Efficiency. The MLET's design minimizes the time required for adjustments, allowing for quicker responses to changes detected by the downhole sensors. This efficiency is critical in maintaining the overall timeline of the operation and reducing non-productive time.



Figure 3—Hydraulic MLET BHA

Data Transmission

Data from the downhole sensors are transmitted to the surface via a tubing encapsulated electrical cable (TEC), ensuring continuous power supply and eliminating the risks associated with downhole batteries. This setup allows for uninterrupted real-time monitoring, enhancing decision-making during operations.

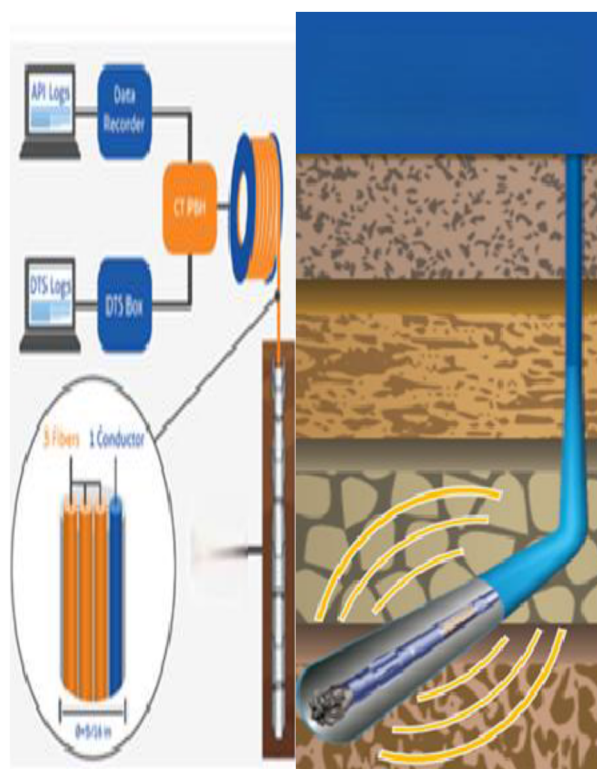


Figure 4—Live Transmission Data Through Coiled Tubing

Full BHA Description

A detailed description of the BHA configuration is as follows:

1. **External CT connector:** allows installation of the BHA to the CT pipe.
2. **Real-Time Downhole Sensors (RTDS):** CCL, Pressure/Temperature, Tension/Compression, Tool Face. Includes check valves, disconnect, circulation sub & burst disc.
3. **Multi-Cycling Valve (MCV):** diverts fluid between inside BHA to annulus.
4. **Multi-Lateral Entry Tool (MLET):** provides indexing and bending capabilities to enter laterals
5. **Nozzle:** enables downhole fluid jetting.

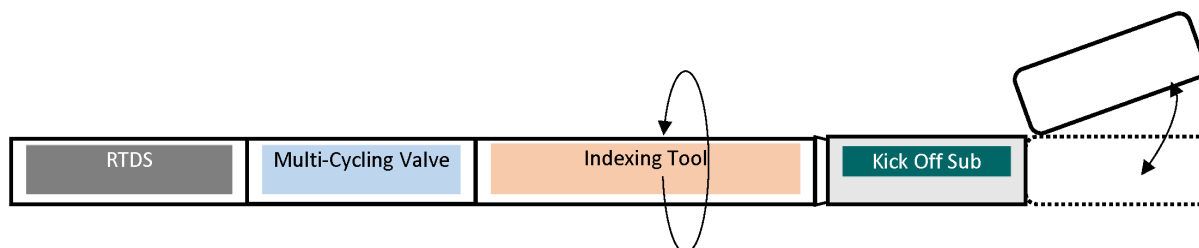


Figure 5—MLET Components with MCV & RTDS

Tool Face Case History

Executive Summary

The dual-lateral acid stimulation treatment intervention executed with precision and success, utilizing advanced downhole technology. The operation employed the MLET in conjunction with RTDS, enabling seamless access to the lateral section and the motherboard of the well. This sophisticated approach ensured optimal reach and effective treatment in the targeted zones.

CT successfully navigated to the L-0-1-1 lateral, confirming its accessibility by detecting a loss of signal on the CCL within the RTDS. This loss of signal was a key indicator of proper tool placement within the lateral, ensuring the treatment would be delivered precisely where needed.

Following this confirmation, the CT team proceeded to carry out the acid stimulation, meticulously working to dissolve and remove near wellbore damage. This critical intervention was performed in both the L-0-1-1 and L-0-1 Mother bore (MB), ensuring comprehensive coverage of the affected areas.

Upon completion of the acid treatment, the CT was pulled out of hole (POOH) to a depth of 3,000 feet. At this period, a soaking period of 4 hours was initiated, allowing sufficient time to react. After the soak, the CT team prepared to initiate the next phase of the operation.

CT then proceeded to drift the ball to open the circulating sub port, marking the transition to the nitrogen (N₂) lifting and flowback operation. This phase was critical for removing the spent acid and residual fluids from the wellbore, facilitating the restoration of flow and well productivity.

During the initial 24 hours of N₂ lifting, the operation successfully brought 90% of oil to the surface, accompanied by water and N₂. This early result indicated a positive response from the well, with further flowback operations ongoing at the time of reporting. The data gathered during this period would be instrumental in assessing the overall success of the intervention and guiding any necessary follow-up actions.

This comprehensive intervention underscores the importance of meticulous planning, precise execution, and close collaboration for this successful operation.

Operation Sequence

The objective of this operation was to perform Dual-lateral acid stimulation through (L-0-1-1 & L-0-1 MB). Function test to real-time downhole telemetry system, MLET and MCV. CT started running in hole (RIH) while pumping minimum rate 0.3 bpm. CT then started mapping ~50 ft above and below L-0-1-1 window, CT managed to access L-0-1-1 lateral with confirmation of CCL lost signal from the second mapping attempt. CT then increased the pumping rate to isolate the MLET and activate the MCV while pumping acid stimulation treatment.

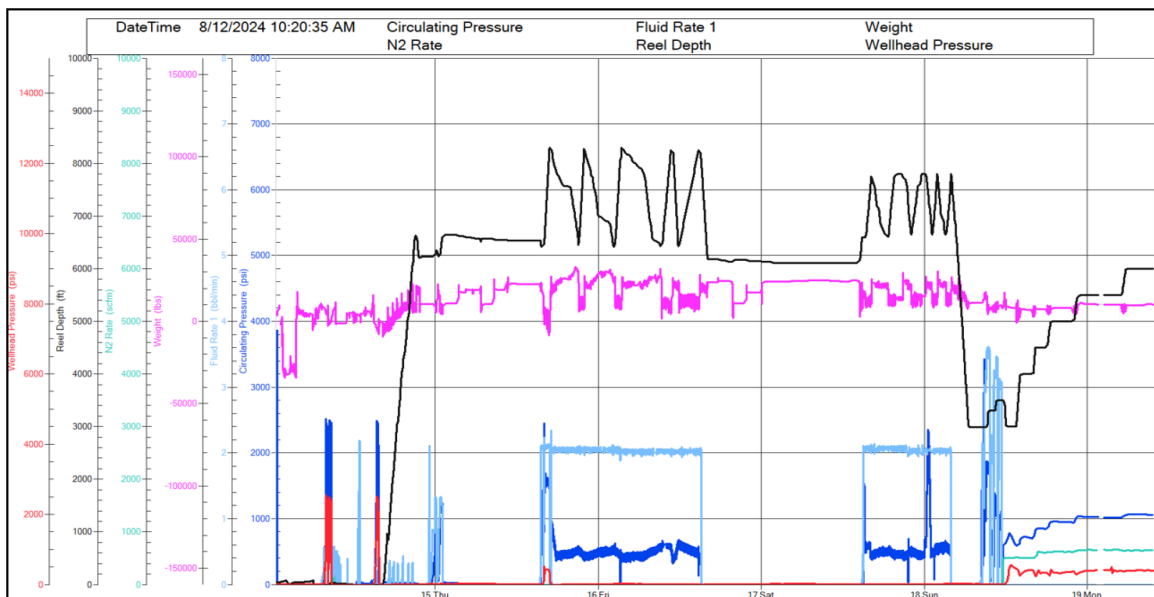


Figure 6—Coiled Tubing (CT) Acid STIM Operation Overview for Both Lateral & MB

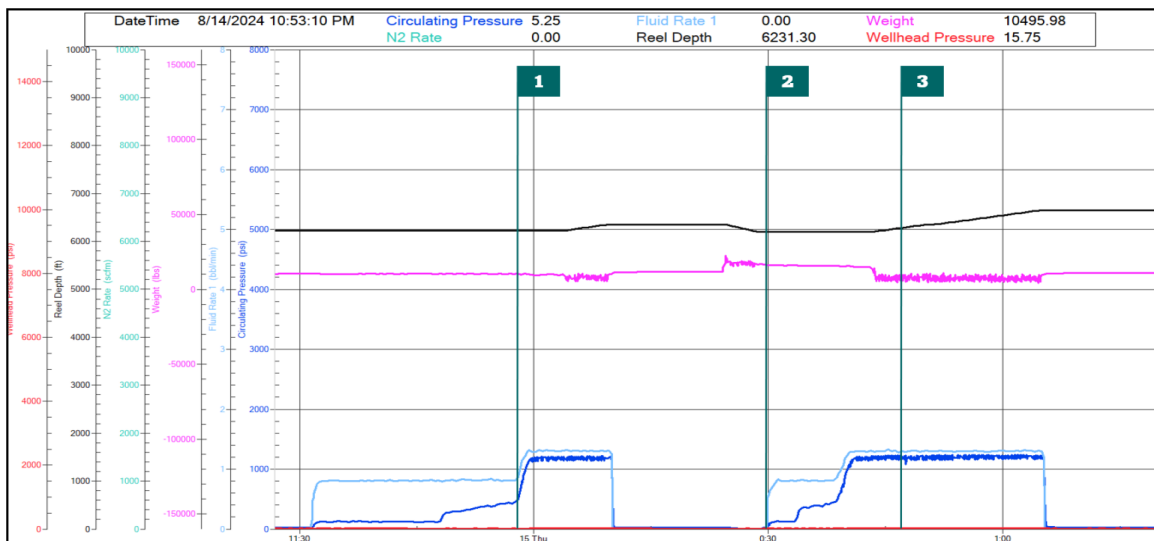


Figure 7—CT Entered L-0-1-1 Lateral Overview

Table 1—Key reservoir parameters of individual layers

Event	Description of Event
1	1 st attempt for CT to map across the window to try accessing L-0-1-1.
2	2 nd attempt for CT to map across the window to try accessing L-0-1-1. (Successful)
3	CT entered L-0-1-1 in 2nd attempt, drop in pressure indication and confirmed accessibility with CCL

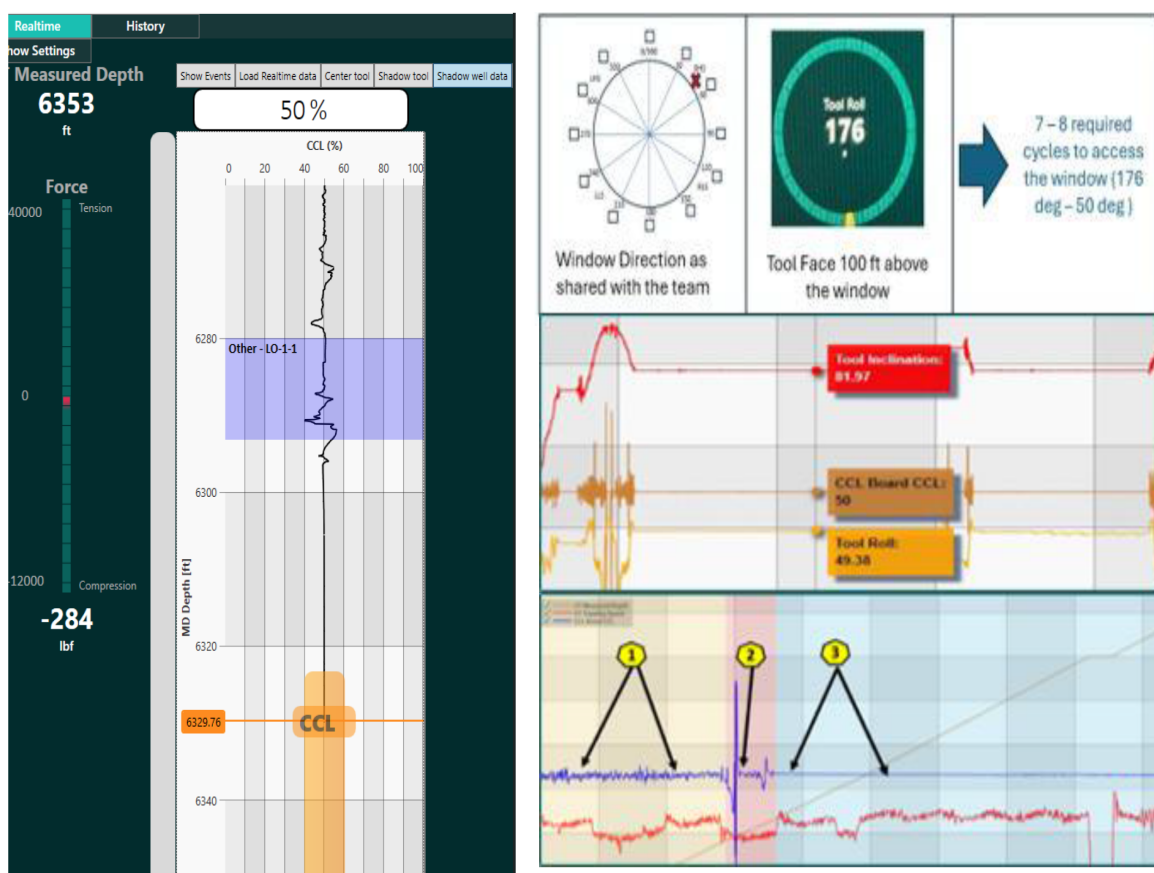


Figure 8—CCL Confirmation & Tool Face of CT entered L-0-1-1 during the second attempt.

Results and Observations

The dual-lateral acid stimulation was executed efficiently, demonstrating the effectiveness of combining MLET with real-time tool face orientation and RTDS. Access to lateral L-0-1-1 was successfully achieved on the second attempt, guided by real-time orientation data and confirmed by the loss of CCL signal, indicating proper tool placement within the lateral. This minimized the number of entry attempts and virtually eliminated the need for conventional mapping procedures, which are typically time-consuming and risk prone.

Acid stimulation was performed in both L-0-1-1 and the mother bore (L-0-1 MB), ensuring thorough treatment coverage for the entire open-hole (OH) section. After CT POOH to 3,000 ft, 4 hours soaking period was observed to allow sufficient Hydrochloric (HCl) acid reaction time. This step was critical for

ensuring the dissolution of near-wellbore damage and maximizing connectivity between the formation and wellbore.

The operation then transitioned to N₂ lifting, which was initiated seamlessly. Within the first 24 hours, the well flowed back to approximately 90% oil, confirming effective acid stimulation and cleanup with strong formation response. The integration of tool face and RTDS not only improved lateral access efficiency but also enhanced operational accuracy, reduced non-productive time, and contributed to a more effective and economical stimulation treatment method.

Conclusions

1. The successful execution of dual-lateral acid stimulation in this operation highlights the significant advantages of integrating tool face orientation technology with RTDS and MLET systems. The ability to monitor and adjust tool orientation in real time was a key enabler in accessing the lateral window efficiently achieving successful entry on the second attempt and nearly eliminating the need for the traditional, time-consuming mapping process.
2. The tool face system provided continuous orientation feedback from downhole, allowing precise alignment of the MLET with the lateral window. This capability not only reduced operational time but also increased confidence in tool deployment and lateral targeting. The reduction in mapping time directly contributed to a shorter intervention window, minimized risk of CT fatigue, and enhanced overall execution reliability.
3. Furthermore, the use of RTDS enabled accurate confirmation of tool placement, such as the loss of CCL signal when entering the lateral, ensuring the acid was delivered exactly as planned. The rapid and productive flowback response 90% oil recovery within 24 hours validated the treatment's effectiveness and confirmed successful wellbore stimulation and cleanup.
4. In addition to technical performance, the integration of these tools enhanced HSE outcomes by reducing rig time, tool handling, and the number of in-hole attempts. Minimizing iterations helped avoid unnecessary wear on the CT string and mitigated risks associated with extended operations in complex open-hole lateral geometries.
5. This case also reinforced the operational value of simulating scenarios with minimal pumping by reducing the number of mapping attempts helping preserve reservoir conditions and providing a reliable workflow for future deployment.

In summary, this approach demonstrated that real-time tool face orientation, when combined with advanced downhole telemetry and hydraulic entry tools, provides a scalable, efficient, and precise solution for complex dual-lateral interventions. It sets a new benchmark in minimizing non-productive time, improving stimulation accuracy, and optimizing operational performance, ultimately contributing to increased productivity and profitability in challenging well environments. Future applications of this technology promise to further enhance operational efficiency and effectiveness, paving the way for continued advancements in the industry and broader deployment in multilateral field development strategies.

Acknowledgment

We would like to express our gratitude to the engineering teams and field personnel who contributed to the successful implementation of advanced sensor technology with tool face and MLET. Their dedication and expertise were instrumental in overcoming the challenges associated with dual-lateral interventions and delivering a safe and efficient operation under demanding conditions.

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Nomenclature

BHA	Bottom Hole Assembly
RTDS	Real-Time Downhole Sensors
HSE	Health & Safety Environment
TEC	Tubing Encapsulated Electrical Cable
MCV	Multi-Cycling Valve
bpm	Barrels per minute
CCL	Casing Collar Locator
CT	Coiled Tubing
HCL	Hydrochloric Acid
MLET	Multi-Lateral Entry Tool
OH	Open Hole
POOH	Pull Out of Hole
RIH	Run In Hole
MB	Mother Bore
TD	Total Depth

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